

5.9 Classwork: Differentiation and Integration Drill (no calculator)

Find the derivative or integral as indicated. Show your working.

— Differentiation —

1. Find $f'(x)$ given $f(x) = 4x^3 - 6x + 7$.

2. Find $\frac{dy}{dx}$ given $y = 5 \sin x - 2 \cos x$.

3. Find $f'(x)$ given $f(x) = 3e^{2x}$.

4. Differentiate $y = 4 \ln x + \frac{3}{x}$.

5. Find $f'(x)$ given $f(x) = \sin(3x + 1)$. *(chain rule)*

6. Find $\frac{dy}{dx}$ given $y = e^{x^2-1}$. *(chain rule)*

7. Find $f'(x)$ given $f(x) = x^2 \ln x$, for $x > 0$. *(product rule)*

— **Integration** —

8. Find $\int (6x^2 + 4x - 1) \, dx$.

9. Find $\int \left(3 \cos x + \frac{1}{x} \right) \, dx$, for $x > 0$.

10. Find $\int 5e^{3x} \, dx$.

11. Find $\int_0^\pi 2 \sin x \, dx$.

12. Find $\int_1^2 \left(\frac{3}{x} + 2x \right) \, dx$.

13. Given that $f'(x) = 4x + e^x$ and $f(0) = 3$, find $f(x)$.

14. Find $f'(x)$ given $f(x) = (2x + 1)e^x$. Hence find $\int (2x + 3)e^x \, dx$. (*product rule, then reverse*)